Power Budget

Team Number: 204
Project Name: **EGR 304 Thing**
Team Member Names: **Alessandro,** Jaxon, Christian, Conner
Version: **1.0**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Pressure Sensor	NPC-120T	+1 - 10V	1	1	1	mA
	Op Amp	MCP6004-I/P	+1.8 - 6V	1	0.1	0.1	mA
	Character LCD	NHD-0216K1Z-FL-YBW	+4.5 - 5.5V	1	159	159	mA
	H-Bridge	FAN8100N	+2.2 to 9V	1	40	40	mA
	DC Air Pump	Adafruit 4699	+4.5 - 24V	1	280	280	mA
	Solenoid Air Valve	Adafruit 4663	+5 - 6V	1	240	240	mA
	5V Linear Voltage Regulator	L7805CV	+7 - 35V	1	8	8	mA
	Curiosity Nano	PIC18F57Q43	+1.8 - 5.5V	1	300	300	mA

+9V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DC Air Pump	Adafruit 4699	+4.5 - 24V	1	280	280	mA
	Solenoid Air Valve	Adafruit 4663	+5 - 6V	1	240	240	mA
						0	mA
					Subtotal	520	mA
					Safety Margin	25%	
					Total Current Required on +9V Rail	650	mA
Regulator	+9V Unregulated Power Su	SMI18-9-V-P5	90 ~ 264 VAC	1	2200	2200	mA
					Total Remaining Current Available on +9V Rail	1550	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Pressure Sensor	NPC-120T	+1 - 10V	1	1	1	mA
	Op Amp	MCP6004-I/P	+1.8 - 6V	1	0.1	0.1	mA
	Character LCD	NHD-0216K1Z-FL-YBW	+4.5 - 5.5V	1	159	159	mA
	H-Bridge	FAN8100N	+2.2 to 9V	1	40	40	mA
	Curiosity Nano	PIC18F57Q43	+1.8 - 5.5V	1	300	300	mA
					Subtotal	500.1	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	625.125	mA
Regulator	+5V Regulator	L7805CV	+7 - 35V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	874.875	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	SMI18-9-V-P5	90 ~ 264 VAC	+9V	2200	2200	mA
Power Rails Connected to External Power Source 1	+9V Rail			1	650	650	mA
	+5V Regulator	LM7805	+7 - 35V	1	625.125	625.125	mA
					Total Remaining Current Available on External Power Source 1	924.875	mA

External Power Source 2	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit

Power Source 2 Selection	Battery	(full part number)	+9V	-9V	500	500	mA
Power Rails Connected to External Power Source 2	-5V Regulator	(full part number)	(range)	1	500	500	mA
					Total Remaining Current Available on External Power Source 2	0	mA

Calculated Battery Life							
	Component Name	**Part Number**	**Supply Voltage Range**		**Capacity (mAh)**	**Required By Regulators**	
	Battery	(full part number)	+12V		500	500	
					Battery Life	1	hours

Notes:

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit